CONCEPT & INVESTMENT BRIEF / UK PILOT

The spa that closes the loop.

A new generation of thermal-wellness facilities, built beside data centres and powered in part by recovered heat.

RECOVER	REUSE	RESTORE
Low-grade heat, upgraded intelligently	Water systems designed for safe circularity	A premium local wellness destination

NORTHBRIDGE

A fictional venture concept. Prepared for exploration, not as an investment offer or a site-specific feasibility study.

October 2026 | United Kingdom

01 / THE PROPOSITION

Turn excess heat into a place people want to be

Northbridge designs, builds and operates premium spa destinations close to data centres, converting a local energy-integration opportunity into a customer-facing wellness business.

The idea in one sentence

Use a carefully contracted supply of data-centre waste heat to reduce the spa's heat demand, combine it with high-efficiency heat pumps and thermal storage, and build a high-quality wellness offer that can trade all year.

01 / INFRASTRUCTURE	**Make useful heat**	Treat recovered heat as an engineered utility, not free energy. Measure temperature, availability and delivered MWh; use heat pumps and storage where the temperature lift is needed.
02 / EXPERIENCE	**Create a destination**	Thermal pools, sauna and steam circuits, treatment rooms, contrast therapy, quiet lounges, café and memberships create reasons to visit beyond a single energy-saving story.
03 / OPERATIONS	**Run it as a system**	Pair wellness operations with metered energy, separate water treatment, predictive maintenance, transparent impact reporting and a clear fallback heat source.

Why Northbridge may be different

- **Developer plus operator:** Northbridge can own and operate a flagship, or deliver a design-build-operate contract for a hotel, campus or property partner.
- **Energy partner by design:** the data-centre operator provides access to an agreed heat source, site interfaces and operational data under a long-term heat-offtake agreement.
- **Community asset:** the spa gives a local, visible use to part of the digital infrastructure economy - potentially supporting staff wellbeing and neighbourhood amenities.

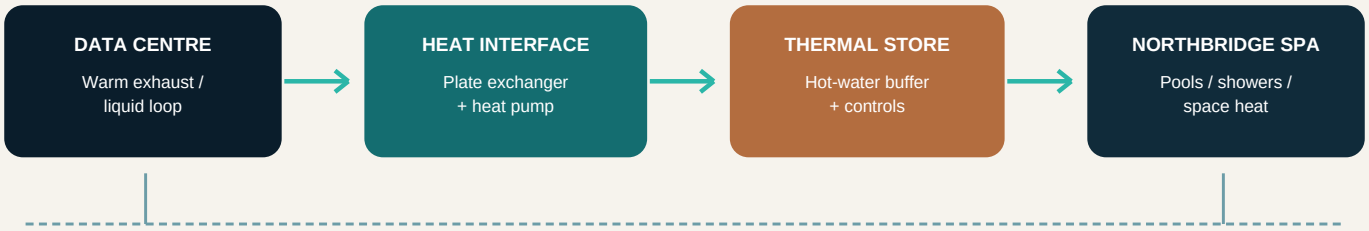
Strategic discipline
A spa is not automatically a good heat offtaker. The business only works where year-round customer demand, heat-source performance, electricity cost, water supply, access and planning align. Validate the site before committing construction capital.

02 / SYSTEMS ARCHITECTURE

Recover heat. Keep every loop controlled.

The data centre and the spa remain independent critical environments. Heat moves across an engineered interface; water quality and public-health controls remain under the spa operator.

CONCEPTUAL ENERGY AND WATER FLOWS



SEPARATE WATER SYSTEM: potable supply + engineered non-potable reuse where permitted; no cross-connection with server cooling loop.

The operating sequence

1. SOURCE	Agree a heat-source envelope: temperatures, hourly and seasonal availability, flow rates, planned outages, water chemistry at the source side and responsibility for maintenance.
2. INTERFACE	Use a plate heat exchanger to isolate circuits. Heat pumps raise the temperature to the level required by pools, domestic hot water and space heating; control systems prioritise the best use of each temperature band.
3. BUFFER	Thermal storage absorbs short-term mismatches between heat supply and demand. Controls avoid depending on a perfect match between server load and spa occupancy.
4. BACKUP	Retain a dependable auxiliary heat source and define safe fallback modes for heat-pump faults, data-centre maintenance or commercial interruptions.

Water is a separate design workstream

Potable supply	Use a compliant potable-water supply for uses that require it, with backflow protection and documented plumbing design.
Non-potable reuse	Assess rainwater and treated greywater for suitable non-potable end uses, such as WC flushing or irrigation, only where design, treatment, monitoring and approvals support it.
Bathing water	Pools, hot tubs and treatment systems need dedicated water treatment, testing, records and operational control. Do not mix data-centre cooling water with bathing water.

Design note: the heat-recovery opportunity is site-specific. The Greater London Authority's 2025 heat-reuse study highlights data-centre cooling technology, utilisation, location, ownership and heat-pump siting as major feasibility factors [1].

03 / THE FACILITY

Premium wellness, not an industrial amenity

Build a strong guest proposition that stands on its own merits. Recovered heat should support the economics and impact story, not substitute for design quality, hospitality or service.

THERMAL CIRCUIT	Warm-water pool, vitality pool, contrast pool, sauna and steam rooms, cold plunge, experience showers and a calm circulation route.
RECOVERY & TREATMENT	Massage and body treatments, relaxation pods, guided breathwork, quiet rooms and bookable recovery sessions for people who work in high-intensity digital and industrial environments.
SOCIAL & HOSPITALITY	A high-quality café, hydration bar, light food, retail essentials, private hire and small corporate wellbeing packages.
MEMBERSHIP	Local monthly membership, off-peak passes, multi-visit bundles and corporate memberships sold to data-centre operators, technology firms, hotels and nearby employers.

Priority customer groups

Local residents Regular wellness and short-break visits.	**Data-centre workforce** Shift-friendly opening, recovery and membership options.
Corporate buyers Team sessions, wellbeing budgets and private bookings.	**Destination guests** Weekend packages with local hotels, restaurants and tourism providers.

Operating principles

- Accessible arrival: safe pedestrian routes, cycling facilities, public transport where available and a clear separation from secure data-centre perimeters.
- Quiet, biophilic interiors: natural materials, restrained lighting, acoustic control and a landscape that hides plant and service functions.
- Measured sustainability: disclose metered recovered heat, heat-pump electricity, water intensity and pool-water compliance - avoid vague “zero-energy” claims.

04 / WHERE IT WORKS

Select for heat, people and reliable access

Northbridge should screen a portfolio of candidate data-centre clusters before selecting a pilot. A site with abundant heat but poor access or limited local demand may be weaker than a smaller, well-connected source.

SCREENING FACTOR	WHAT TO CONFIRM	RED FLAG
Heat quality and quantity	Hourly heat profile, supply temperature, source loop, recoverable MW, seasonality, maintenance calendar and metered evidence.	Only theoretical heat; no operating data or no agreed offtake.
Distance and route	Practical pipe route, land rights, crossings, insulation, pumping load and room for an energy centre.	Major road/rail/river crossings or a long expensive route.
Guest market	Population and employers within drive/transit catchment; competitor pricing; hotel partners; weekday/weekend demand.	Demand depends on the data-centre workforce alone.
Water and drainage	Potable-water capacity, discharge limits, treatment plant, water scarcity, stormwater and reuse opportunities.	Unresolved discharge, supply or drought resilience.
Planning and neighbours	Use class, traffic, noise, lighting, ecology, flood risk, operating hours, visual design and community feedback.	Planning depends on assumptions not yet tested.
Commercial structure	Heat price formula, minimum availability, outages, termination rights, ownership of plant, performance tests and insurance.	No fallback supply or one-sided termination rights.

Candidate archetypes

Urban edge A compact premium spa near a colocation or enterprise data-centre cluster, with strong public transport and dense local demand.	**Technology campus** A larger recovery and wellness venue paired with several employers, with corporate memberships and weekday demand.	**Regeneration district** A mixed-use redevelopment where heat infrastructure, hotels, housing and leisure can share a wider energy strategy.

Pilot rule: commission independent energy and market diligence in parallel. Do not acquire a site or order long-lead plant until the heat-offtake terms, network route and planning path are credible.

05 / ILLUSTRATIVE UNIT ECONOMICS

A pilot business, with the assumptions in view

The following is a worked example for a medium-sized, premium UK spa. It is not based on a named site, operator quote, customer survey or verified heat profile. All figures are indicative and exclude VAT unless noted.

95,000
paid visits / year

£7.54m
annual revenue

£1.77m
illustrative EBITDA

£18m
midpoint build cost

REVENUE ASSUMPTION	CALCULATION / BASIS	ANNUAL
Admission and day passes	95,000 visits with £52 blended admission yield	**£4.94m**
Treatments	Treatment menu, utilisation and therapist capacity	**£0.95m**
Food, beverage and retail	Net sales contribution, with direct product costs included below	**£0.75m**
Memberships and corporate	Recurring plans, employer packages and private sessions	**£0.70m**
Events and other	Small private hires and ancillary revenue	**£0.20m**
TOTAL REVENUE	Illustrative stabilised year	**£7.54m**

OPERATING COST	ANNUAL
Staff, training and employment costs	£2.25m
Energy, water and heat-network charges	£0.55m
F&B; and treatment consumables	£0.62m
Property, insurance and maintenance	£0.85m
Marketing, technology and administration	£0.80m
Other costs and operating contingency	£0.70m
TOTAL OPERATING COST	**£5.77m**
ILLUSTRATIVE EBITDA	**£1.77m**

BUILD COST / CAPITAL NOTE

Concept range: £15m-£25m for a leased or brownfield site, including building works, wet leisure fit-out, plant, professional fees and contingency. The £18m midpoint is illustrative. Land purchase, VAT, financing, major off-site heat-network works and exceptional ground conditions are excluded.

EBITDA margin: c. 23.5% in this example. EBITDA is before rent/finance if not captured in property costs, depreciation, tax, debt service, lifecycle capex and ramp-up losses. It is not a project IRR or equity return.

Sensitivity matters more than the headline

If revenue lands 20% below plan while the cost base only partly flexes, EBITDA can collapse. The investment case should be stress-tested for visit volumes, average spend, wage inflation, electricity prices, plant uptime, heat tariff, capex overruns and the time needed to reach stabilised occupancy.

06 / HOW NORTHBRIDGE SCALES

A repeatable delivery and operating model

Start with one reference site and standardise the parts that repeat: contract structure, heat interface, metering, spa operating procedures, water-quality systems and impact reporting.

MODEL	NORTHBRIDGE ROLE	BEST FIT
Own and operate	Northbridge develops or leases the asset, invests capital, employs the team and keeps operating upside and risk.	Flagship site; strong local market; long-term site control.
Design-build-operate	A property or leisure owner funds the asset; Northbridge designs the concept, integrates plant and operates under a management agreement.	Hotel, regeneration project or institutional asset owner.
Joint venture	Data-centre / property / capital partners share development cost, governance and economics under a defined risk allocation.	Large campus with a committed heat and land partner.

Suggested 18-month development roadmap

0-3 MONTHS	**ORIGINATION**	Shortlist 3-5 sites; introductory discussions with operators, local authorities, heat-network developers and property owners; agree data-request template.
3-6 MONTHS	**FEASIBILITY**	Metered heat study, market and competitor review, concept design, route and water review, preliminary planning advice and order-of-magnitude cost plan.
6-9 MONTHS	**CONTRACT & DESIGN**	Heads of terms for heat offtake and land, front-end engineering, operator model, public-health strategy, planning submission and funding plan.
9-18 MONTHS	**DELIVER & COMMISSION**	Detailed design, procurement, construction, staff recruitment, commissioning, water-safety controls, performance tests and soft launch.

Contract principles to protect the business

- Heat offtake agreement: source-side temperature and availability bands, metering, price formula, outage notice, access, liability, change control and end-of-term rights.
- Operational separation: spa service must not be able to compromise the data centre's cooling, resilience, security or uptime. Design the recovery interface so either party can isolate it safely.
- Step-in and exit: include a credible auxiliary heat strategy and a commercial path if the data centre changes cooling design, expands, relocates or ceases operations.

07 / CONTROLS AND NEXT DECISIONS

Build trust into the design

The concept combines leisure, public health, building services and energy infrastructure. These workstreams need named owners from feasibility onward.

RISK AREA	NORTHBRIDGE RESPONSE
Legionella and spa-pool water	Create a documented water-safety plan; competent design and operations; risk assessment, cleaning, sampling, monitoring, records and appropriate staff training. HSE HSG282 is a key starting point for spa-pool systems [2].
Heat intermittency / failure	Model hourly supply and demand; install buffer storage and backup heat; set clear reliability and outage terms; test fallback operating procedures.
Water scarcity and discharge	Complete a water balance; use efficient fixtures and recovery; confirm discharge and treatment requirements; reuse only where safe, lawful and technically justified.
Planning / community	Engage early on traffic, noise, lighting, opening hours, landscape, ecology and access. Keep all guest-facing design separate from restricted data-centre operations.
Regulatory classification	Obtain specialist advice on planning, building regulations, workplace and pool safety, water and environmental permits, fire strategy, consumer terms and any heat-network obligations.
Demand / capex risk	Use customer research and pre-sales; benchmark comparable venues; independently cost the project; add contingency and phase discretionary features.

Regulatory watch

In Great Britain, heat networks are being brought into a phased regulatory framework overseen by Ofgem. Government guidance updated in 2026 describes a Heat Network Technical Assurance Scheme, with planned launch in 2027. Whether a particular Northbridge arrangement falls within heat-network rules depends on its structure and supply relationships; obtain project-specific advice rather than assuming every on-site recovery loop is regulated in the same way [3][4].

Recommended next decision

Authorise a limited feasibility phase only: obtain site-specific hourly heat data, a heat-offtake term sheet, independent spa market study, water/energy concept design and a quantity-surveyor cost plan. Proceed to full development only when the combined technical, planning and commercial gates are met.

Selected references

[1] Greater London Authority / AECOM and HermeticaBlack, *Optimising Data Centres in London - Heat Reuse*, report issued for upload 22 April 2025.
https://www.london.gov.uk/sites/default/files/2025-06/Optimising_Data_Centres_in_London_-_Heat_Reuse_250605.pdf

[2] UK Health and Safety Executive, *HSG282: Control of legionella and other infectious agents in spa-pool systems*. Online guidance page, updated 7 August 2025.
https://www.hse.gov.uk/PuBns/books/hsg282.htm

[3] Ofgem, *Heat networks* - regulatory overview and consumer protections.
https://www.ofgem.gov.uk/low-carbon/heat-networks

[4] UK Department for Energy Security and Net Zero, *Heat Network Technical Assurance Scheme (HNTAS)*, updated 16 February 2026.
https://www.gov.uk/government/publications/heat-networks-regulation-technical-standards/heat-network-technical-assurance-scheme-hntas

[5] UK Parliament, written answer on integrating data centres into heat-network zoning plans, answered 22 May 2026.
https://questions-statements.parliament.uk/written-questions/detail/2026-05-13/1035/

Important: This document is a fictional business concept and indicative worked example, prepared in October 2026. It is not engineering advice, legal advice, a planning opinion, a financing proposal or an offer to invest. Values are assumptions to be validated by competent professionals and site-specific evidence.